

A prototype coupled modeling approach for predicting harmful algal blooms: A case study in Chile

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ARTICLE INFO

Keywords:

Bloom forecasting
Empirical dynamic modeling
Nonlinear time-series analysis
Pseudo-nitzschia
Causal species

ABSTRACT

Predicting harmful algal blooms (HABs) remains a major challenge for coastal management and aquaculture. This study compares three forecasting approaches developed under the Monitoring of Algae in Chile (MACH) project: a particle dispersion model, an LSTM neural network, and an empirical dynamic model (EDM) to evaluate their ability to forecast bloom events. Consequently, we applied the EDM to forecast two *Pseudo-nitzschia* species groups using data collected from Metri, Quellón, and Melinka in southern Chile. The results showed that the genus *Ceratium* and *Leptocylindrus* were commonly associated with both *Pseudo-nitzschia* species groups, and the best prediction by causal species was obtained for the *P. seriata* group, with a correlation coefficient of 0.733 ($P < 0.0001$) between observed and predicted values. This case study demonstrated that species interactions can be used to predict specific HAB species; however, the prediction performance may vary depending on location and species. This study provides one of the first applications of EDM for HAB forecasting using causal species in a real-world monitoring context, demonstrating the potential of hybrid modeling frameworks to improve early warning systems and mitigate aquaculture losses.

Abbreviations: AI, Artificial Intelligence; ANNs, Artificial neural networks; CCM, convergent cross mapping; chl-a, Chlorophyll-a; CROCO, The Coastal and Regional Ocean Community Model; EDM, Empirical dynamic model; HAB, Harmful algal bloom; IFOP, Instituto de Fomento Pesquero; INTESAL, Instituto Tecnológico del Salmón; LSTM, Long Short-Term Memory; MACH, Monitoring of Algae in Chile; MAE, Mean absolute error; Parti-MOSA, The particle-dispersion model; PROMOFI, The Phytoplankton Monitoring Program; RMSE, Root mean squared error; SATREPS, Science and Technology Research Partnership for Sustainable Development.

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<https://doi.org/10.1016/j.ecoinf.2026.103615>

Received 20 May 2025; Received in revised form 14 January 2026; Accepted 14 January 2026

Available online 16 January 2026

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1. Introduction

Predicting the location, timing, and magnitude of harmful algal blooms (HABs) is critical for coastal fisheries, government agencies, and the general public, as it helps protect fish and shellfish farms and ensures seafood safety for human consumption (Anderson et al., 2012; Díaz-Tapia et al., 2019; Hallegraeff et al., 2021; Lassudrie et al., 2020). However, achieving accurate HAB predictions is challenging due to the complex nature of HAB species, which differ in morphology, life cycle, toxicity, and their responses to environmental conditions (Garcés et al., 2001). Despite global progress in identifying bloom-forming algae and understanding HAB dynamics, prediction remains difficult. Advanced technologies have improved coastal monitoring, species identification, and the measurement of oceanographic parameters involved in HAB formation (Anderson et al., 2012). Yet, the frequency of sampling is often insufficient to support reliable prediction models, as algae are constantly moving and subject to significant spatial and temporal variability. Therefore, improved modeling approaches are urgently needed to integrate available data and provide timely, accurate HAB warnings. This need is especially relevant to fish and shellfish farmers and fishermen, who face financial risks from unexpected closures of farms and shellfish beds. While both groups are affected, farmers are more likely to benefit from short-term HAB forecasts, as the ability to plan and close fish cages in advance of a bloom can protect their stock and increase profitability (Trotter et al., 2018, 2022). Rather than long-term predictions, 1- to 2-week forecasts are more practical, allowing farmers to harvest their products before a bloom occurs. However, there is a trade-off: false positive predictions can lead to premature harvesting and loss of revenue. To address this, HAB models must be optimized to minimize prediction errors and avoid unnecessary closures. Accurate short-term forecasts not only help farmers make informed decisions but also allow unaffected farms and shellfish beds to remain open, preserving income and reducing waste (Jin and Hoagland, 2008).

Chile is the world's second-largest producer of salmon and one of the top global exporters of frozen mussels from marine farms. Despite its importance in the aquaculture industry, southern Chile has been severely affected by HABs over the past few decades (Díaz-Tapia et al., 2019). In 2016, a major HAB event involving *Pseudochattonella verruculosa* affected 18–20% of Chilean salmon production, resulting in losses of approximately USD 800 million. Additionally, *Alexandrium catenella* blooms caused an estimated USD 30–40 million in losses to mollusk production during the austral summer of 2016 (Díaz-Tapia et al., 2019).

Chilean salmon farms are primarily located in northern and central Patagonia (southern Chile), home to the country's most extensive fjord systems (Pantoja et al., 2011), while mussel farms are concentrated in the northern area of these fjords. The ecosystem in this region is characterized by high salinity in deeper waters and low salinity in surface waters (Iriarte et al., 2010). It also exhibits high environmental heterogeneity, influenced by its complex physical morphology, river discharge, glacial melt, rainfall patterns, wind speed and direction, and solar radiation (Calvete and Sobarzo, 2011; Pérez-Santos et al., 2014; Pinilla et al., 2020; Saldías et al., 2019; Sievers, 2008; Strub et al., 2019). These contrasting conditions lead to significant variability in the physico-chemical and biological properties of the water column. Therefore, routine monitoring and the development of ecological models are essential to better understand this dynamic environment and improve HAB prediction capabilities.

The goal of this paper is to advance understanding and predictive capacity for HABs by both synthesizing existing modeling approaches and introducing novel predictive frameworks. Specifically, we (1) provide a comprehensive review of current HAB prediction and forecasting models, categorizing them into empirical-statistical, mechanistic, and physical approaches; and (2) present and evaluate three new models developed under the SATREPS–MACH project (Science and Technology Research Partnership for Sustainable Development – Monitoring of Algae in Chile). These include: (a) a physical model (Parti-MOSA), (b) an

artificial intelligence model based on Long Short-Term Memory (LSTM) networks that integrates holobiome monitoring via DNA metabarcoding, and (c) an advanced Empirical Dynamic Model (EDM) applied to long-term community data. Together, these models address critical gaps in forecasting HAB dynamics in the highly complex and understudied Chilean Patagonian environment. As a demonstration, the EDM approach is applied in a case study to predict two harmful *Pseudonitzschia* species groups using long-term phytoplankton monitoring data.

2. Types of HAB prediction models

An optimal HAB prediction model can provide quantitative estimates of the likelihood, intensity, and movement of a HAB across spatial and temporal gradients. Such a system plays a crucial role in minimizing the economic and health-related impacts of HABs. An effective alert system typically relies on one or more types of models or their combinations. According to Anderson et al. (2015), HAB prediction models can be broadly categorized into three types: empirical-statistical, mechanistic, and physical.

2.1. Empirical-statistical models

Research publications on HAB forecasting have increased significantly with advances in monitoring technologies (Rouso et al., 2020). Data-driven forecasting models, particularly empirical-statistical models, have received considerable attention due to modern machine learning (ML) tools. These models range from simple correlations, time lags, and pattern recognition to sophisticated neural networks and fuzzy logic (Anderson et al., 2015).

However, empirical-statistical models have limitations. They are constrained by training datasets, making extrapolation to novel conditions difficult, such as HABs driven by anthropogenic factors like eutrophication. These models often predict single variables (e.g., chlorophyll-a, algal density) based on environmental factors. According to Rouso et al. (2020), the number and sophistication of empirical-statistical models have grown over time. Machine learning techniques applied to HAB prediction have shown strong performance across lakes and coastal systems with varying eutrophication levels. Since early work by Recknagel et al. (1997) using artificial neural networks (ANNs) for complex, nonlinear algal dynamics, artificial intelligence has played an increasingly important role in HAB forecasting.

Recent methods include Long Short-Term Memory (LSTM) networks (Hill et al., 2020; Liang et al., 2020; Zheng et al., 2021), random forests (Derot et al., 2020; Segura et al., 2017), support vector machines (Vilas et al., 2014), and gradient boosting machines (Jeong et al., 2022; Xia et al., 2020). Model performance is influenced by uncertainties in model structures and input data errors (Busari et al., 2024). In many HAB prediction studies, chlorophyll-a is widely used as an indicator of eutrophication. Performance is typically assessed using statistical metrics such as mean absolute error (MAE), coefficient of determination (R^2), and root mean square error (RMSE) (Alsini et al., 2025; Mozo et al., 2022). Table 1 summarizes the main modeling techniques used in empirical-statistical models, with representative examples for each type. Despite their predictive accuracy, these models typically rely solely on environmental variables as predictors and do not explicitly incorporate biological interactions between species, limiting their ability to capture ecological dynamics driving HAB events.

2.2. Mechanistic

A mechanistic model is developed by numerically parameterizing the fundamental physiological characteristics of a target organism. This approach is less commonly used because it requires detailed knowledge of the organism's ecology, including environmental niche, life history traits, and toxin production, which can limit applicability to specific species and regions. In community ecology, the focus is on

understanding how species are distributed across space and time and identifying factors that shape community structure (Leibold et al., 2004; Vellend, 2010; Weiher et al., 2011). While broad theoretical frameworks, such as Margalef's mandala (Margalef, 1978) and the C-S-R model (Reynolds, 1984), offer insights into global algal community assembly (Reynolds, 2006; Smayda and Reynolds, 2001; Wyatt, 2014), understanding HAB dynamics requires investigation of local processes, particularly in intensively managed aquatic systems. Examples of mechanical models include deterministic models for *Alexandrium fundyense* bloom dynamics (He et al., 2008; McGillicuddy et al., 2011; McGillicuddy Jr et al., 2005; Stock et al., 2005), *Karenia brevis* (Olascoaga et al., 2008), and other bloom-forming dinoflagellates (Chakraborty et al., 2022; Chakraborty and Feudel, 2014).

2.3. Physical models

Physical models, such as hydrodynamic circulation models, are used to study bloom formation, duration, and physical processes like upwelling (Palma et al., 2010; Sacau-Cuadrado et al., 2003) and wind-driven transport (Raine et al., 2010) that primarily influence harmful algal species dispersion. Because these models are not dependent on specific algal types, they are versatile and often demonstrate high accuracy in short-term predictions. For example, a particle-tracking model

examining a *Chattonella* bloom in the Yatsushiro Sea identified wind and river discharge as key drivers (Aoki et al., 2012). Similarly, Li et al. (2009) developed a Lagrangian particle simulation in the Gulf of Maine, later extended by Henrichs et al. (2015) to include biological behavior such as vertical migration of *Karenia brevis* cells. Aleynik et al. (2016) introduced an advanced hydrodynamic model for coastal waters off western Scotland, employing an unstructured finite volume approach with high spatial resolution, enabling realistic simulations of particle transport such as *Karenia mikimotoi* cells within nearshore aquaculture zones. When well-developed, physical models can be integrated into management frameworks to support decision-making and help mitigate HAB health and economic impacts. The particle-dispersion model (Parti-MOSA) employed in this study represents such a physical modeling approach for HAB forecasting in Chilean waters.

2.4. Coupled models for HAB forecasting

Coupled modeling approaches use multiple models to evaluate how different inputs affect predictions. Given HAB complexity, single models may be insufficient. Coupled approaches integrate diverse data sources and modeling assumptions, such as ecosystem parameters, climate variability, and stochastic physical-biological interactions, to improve robustness (Ralston and Moore, 2020). Publications have increased

Table 1

Brief comparison of empirical-statistical approaches in HAB model building.

Modeling technique	Key features	Applicability	Complexity	Accuracy	Several references for applications on HABs
Regression	- Techniques such as regression analysis analyzes historical datasets to forecast the algal bloom. - Predicts the effect of an independent variable on the dependent variables. - Linear regression, logistic regression, support vector machines, multiple adaptive regression splines are some of the varieties of regression analysis.	Well suited for linear systems.	Less complex.	- A strong correlation does not imply cause and effect relationships. - Could have low performance. - Better for only short term prediction.	Melendez-Pastor et al., 2019 Mchau et al., 2019 Richardson et al., 2018
Artificial neural networks	- Computer learns to perform some tasks by analyzing training examples. With enough training data, Artificial neural networks can generalize and transfer the learned relations to new data. - Convolutional neural networks, Recurrent neural networks, Long short-term memory are some of the varieties of neural network analysis.	- Well suited for dynamic non-linear systems. - Seen in many real-world problems.	- Highly complex with a steep learning curve. - Hard to interpret due to its "black box" nature.	Comparatively accuracy is high but depends on training data.	Wen et al., 2022 Cruz et al., 2022 Chen et al., 2025
Decision trees	- In a decision tree, the input data are repeatedly partitioned into two groups based on certain criteria until a predetermined stopping condition is met. - Classification trees, Random forest, Model trees, Bagged trees, Extra trees are some of the model variations of decision trees.	- Suited for non-linear systems - Data pre-processing needs less effort.	Easy to understand and interpret.	Comparatively accuracy is high but is prone to overfitting	Harris and Graham, 2017 Harley et al., 2020
Bayesian networks	A probabilistic modeling approach which includes linear, nonlinear, combinatorial and stochastic predictive relationships across multiple variables.	- Applicable for both qualitative and quantitative data and can accommodate data gaps. - Less requirements for the sample size.	- Computationally demanding for systems with a large number of random variables. - Requires substantial effort/expertise to construct for complex systems.	Comparatively accuracy is high.	Wang et al., 2022 Liu et al., 2023
Fuzzy logic	Fuzzy logic is an approach to variable processing that allows for multiple possible truth values to be processed through the same variable.	Suited for non-linear systems.	The structure of Fuzzy Logic system is simple to understand.	Comparatively accuracy is high but depends on setting accurate fuzzy guidelines.	Chen and Mynett, 2003 Yang et al., 2021
Empirical dynamic modeling	Non-parametric framework ("equation-free") for understanding and predicting ecosystem behavior when linear statistics and parametric models do not apply.	Suited for non-linear systems.	- Highly complex with a steep learning curve. - Computationally demanding.	Prediction accuracy is constrained by the quantity and quality of data.	McGowan et al., 2017 Baydaroglu, 2025 Tian et al., 2024

exponentially since 2000, demonstrating the field's rapid maturation (Supp. Fig. 1). Table 2 summarizes recent coupled modeling approaches (2023–2025).

Coupling physical and biological models in HAB studies dates back to Franks (1997), who demonstrated their importance in understanding HAB ecology. Since then, real-time predictions have combined satellite observations with ecosystem models to enhance predictive capabilities. A widely used example is the NPZD (Nitrogen-based nutrients, Phytoplankton, Zooplankton, and Detritus) ecosystem model (Luo et al., 2012; Macdonald and Gulbudak, 2023; Olascoaga et al., 2005; Tian et al., 2015), often linked with hydrodynamic models (Duquesne et al., 2021; Liu et al., 2025; Omlin et al., 2001; Sharip et al., 2016; Weng et al., 2023; Zhao et al., 2024). Operational management systems have also been developed, such as iLake software, which integrates a two-dimensional hydrodynamic module, a mass-transport module, and a phytoplankton kinetics module (Huang et al., 2015). These coupled systems are typically operated by governing agencies to protect public health and inform coastal industries (Franks, 2018). However, many models estimate total chlorophyll biomass rather than species-specific abundances (e.g., Allen et al., 2008), highlighting a critical gap in modeling specific HAB species dynamics.

Many of the recently published studies on ML-coupled models use different modeling approaches and evaluate each model's efficiency in prediction rather than a coupled approach. This ML model ensemble approach integrates several ML techniques to create a combined prediction model (Ahn et al., 2023). For instance, Szewczyk et al. (2025) successfully predicted four HAB species (*Pseudo-nitzschia* sp., *Alexandrium* sp., *Dinophysis* sp., and *Karenia mikimotoi*) using a stacking ensemble meta-model that aggregated six candidate ML models, such as logistic ridge regression, multivariate adaptive regression splines, random forest, extreme gradient boosting, multilayer perceptron, and hierarchical Bayesian logistic regression. Their results demonstrated that ensemble modeling leverages the strengths of various techniques for more accurate and reliable forecasts.

Despite these advances, coupled models face several limitations. They require substantial computational resources and expertise to develop and maintain, often making them impractical for routine operational forecasting. Furthermore, coupled models typically require extensive parameterization and calibration using site-specific data, limiting their transferability across different regions. Most critically, while coupled physical-biological models have advanced our understanding of environmental drivers, they still largely overlook biological interactions between species that may play crucial roles in HAB dynamics. Given the limitations of existing approaches, there is a growing need for complementary methods that can capture species interactions

and provide accurate forecasts with lower computational demands. HAB research has traditionally focused on investigating individual species under controlled laboratory conditions to determine factors triggering their growth and toxicity mechanisms (Chang et al., 2022; Orr and Jones, 1998; Wang et al., 2022). However, this approach often overlooks complex interactions in natural environments and fails to fully capture HAB ecological dynamics. A systematic review by Li et al. (2025) found that of 362 HAB studies, approximately 64% focused on a single species. In marine environments, the three most frequently studied algal genera were *Alexandrium* spp. (25%), *Karenia* spp. (9%), and *Prorocentrum* spp. (9%), with the remaining 57% distributed among other species. The study highlighted a significant gap between laboratory findings and field observations (Li et al., 2025).

HABs emerge within complex ecosystems where numerous bloom-forming and non-bloom-forming algal species, along with various non-algal organisms, interact. These interactions enable certain bloom-forming species to dominate the community, accumulating biomass beyond that of other species. Zhang et al. (2023) demonstrated the importance of integrated approaches in HAB studies, showing that interspecific competition, particularly between diatoms and dinoflagellates, shapes bloom characteristics and outcomes. Their study found that when nitrogen and silica are limited, cyanobacteria and diatoms are outcompeted, allowing dinoflagellates, especially toxin-producing species such as saxitoxin producers, to dominate due to their better adaptation to nutrient-poor conditions. Therefore, our study introduces a novel integrative approach by identifying causal interactions between HAB-forming species and non-bloom-forming algal species and incorporating these interactions into HAB prediction.

3. SATREPS - MACH project prediction models in Chile

The SATREPS-MACH project was a Japan-Chile collaborative research initiative focused on HABs in Chile, aiming to develop a reliable HAB forecasting system from the holobiome perspective to minimize impacts on public health and aquaculture (Fujiyoshi et al., 2023; Gajardo et al., 2023; Yarimizu et al., 2020). The holobiome concept, which is the total genomic content of a eukaryotic organism and its associated microorganisms, is particularly relevant as HABs may be influenced by their associated microbiome.

In Chile, particularly in the Chiloé Island region with intensive salmon and mussel aquaculture, HABs remain a problem that receives limited predictive attention; current efforts largely rely on continuous microscopy-based monitoring by aquaculture industry rather than true forecasting. As HAB frequency has increased in this area, the MACH project investigated HAB species composition, bacterial communities, and local hydrodynamic conditions. The project's objectives were to: (1) identify microorganisms constituting algal blooms (the HAB holobiome); (2) determine factors shaping the HAB holobiome; (3) develop forecast models using HAB species and associated biotic and abiotic factors; and (4) establish a HAB information-sharing network.

This report focuses on the third objective, introducing three HAB prediction models developed within the project: two empirical-statistical approaches using LSTM and EDM, and one physical modeling approach using Parti-MOSA. Their performance was evaluated, with particular emphasis on a case study applying the EDM approach to long-term data from Chilean Patagonia. Finally, the study discusses how integrating these models could enhance HAB forecasting (Supp. Table 1).

3.1. Model 1: Parti-MOSA

Model 1 is a physical/oceanographic model that simulates HAB cell transport via ocean currents without accounting for biological growth or behavior. The South-Austral Operational Particles Forecast Model (Parti-MOSA, from its Spanish acronym) is a particle-dispersion tool based on the MOSA model (Ruiz et al., 2021) and the OpenDrift

Table 2
Coupled HAB prediction model types in the recent past literature.

Model types	Coupling approaches	References
ML coupled with physical models	Ensemble ML models + physical data integration	Lin et al., 2022; Bagherian et al., 2024; Mermer et al., 2025; Ahn et al., 2023
	Bagging and stacking ensemble techniques	
	Linear models + non-linear models	Szewczyk et al., 2025
	Process understanding models + predictive ML classification	Tal et al., 2024; Tian et al., 2024
Hydrodynamic-ecological models	Hydrological models + ecological models	Weng et al., 2023; Zhao et al., 2024; Liu et al., 2025
	Particle tracking models + ecological models	Xiong et al., 2023
Biogeochemical-physical models	Biogeochemical cycles + population dynamics	Macdonald and Gulbudak, 2023
	Satellite observations + Hydrodynamic model + Ecological models	Wu et al., 2023

numerical framework (Dagestad et al., 2018). MOSA is an operational oceanographic and atmospheric forecast model configured for Northern Patagonia, Chile, built upon the Coastal and Regional Ocean Community model (CROCO; Debreu et al., 2012), providing data on water temperature, salinity, sea level, and ocean currents at approximately 1 km horizontal resolution. CROCO is forced by the WRF atmospheric model (Skamarock et al., 2008), which supplies air temperature, heat fluxes, precipitation, and wind speed at 3 km resolution.

With information from MOSA, Parti-MOSA calculates particle drift trajectories (simulating harmful algae) online using instantaneous advection-diffusion equations. Users can define parameters such as release location, depth range, radius, start date, and forecast horizon (3–10 days). The particles behave as neutrally buoyant drifters (Reche et al., 2021). Details are published in Ruiz et al. (2021), and forecasts are available at <https://chonos.ifop.cl/mosa/> and <https://chonos.ifop.cl/partimosa/>. Additionally, ongoing HAB monitoring by Instituto de Fomento Pesquero (IFOP) produces data on relative abundances of seven harmful species based on long-term microscopic observations. When relative abundance reaches a critical threshold (early warning), the Parti-MOSA website generates a particle-tracking simulation to define the potentially affected geographic area (Fig. 1). Although the current model does not yet incorporate species-specific biological parameters, future developments will tailor the model to simulate individual HAB species behavior more accurately. Despite not yet incorporating biological characteristics, this model is widely used by the salmon farming industry during alerts of high abundance of harmful microalgae, helping to support early decision making to prevent mortalities and implement preventive closures of salmon farming sites. Additionally, this dispersion model has been used to reconstruct events that impacted the salmon farming industry. Mardones et al. (2023) analyzed the *Heterosigma akashiwo* bloom in Comau Fjord in Chile (March - April 2021) using climatic conditions, in situ observations, and satellite data, and applied a Parti-MOSA based particle dispersion model

to characterize circulation and water mass retention associated with the observed salmon mortalities.

3.2. Model 2: LSTM artificial intelligence (AI) model based on time series of holobiome monitoring by DNA metabarcoding data

Model 2 is an LSTM neural network, a type of deep learning model capable of learning from time-series data through its internal memory cells and gating mechanism (Hochreiter and Schmidhuber, 1997). In the MACH project, the LSTM model was trained on time-series monitoring data, including DNA metabarcoding of plankton and environmental parameters, to detect and predict the presence of micro-eukaryotic species (including HAB species) based solely on environmental conditions. The model was applied to detect predictable micro-eukaryotic species, including HAB species, using a dataset from Metri (Los Lagos region, Chile) consisting of 135 samples (Fig. 2). Approximately 60–70% of the data was used for training, with the remainder reserved for testing. Preliminary results (to be reported separately) indicate that the model can accurately predict a subset of species, 34 out of 1034 operational taxonomic units (OTUs), including five HAB species.

3.3. Model 3: EDM

The long-term database maintained by the Instituto Tecnológico del Salmón (INTESAL) includes phytoplankton count data enumerated through optical microscopy. This phytoplankton abundance data was used in EDM to explore causal relationships between a target HAB genus, *Pseudo-nitzschia*, and other phytoplankton species (*Other phytoplankton* refers to all phytoplankton species excluding the two target *Pseudo-nitzschia* species groups).

EDM does not assume a fixed functional form, making it particularly effective for analyzing complex ecological data where linear or parametric models may fail (Sugihara et al., 2012; Sugihara and May, 1990).

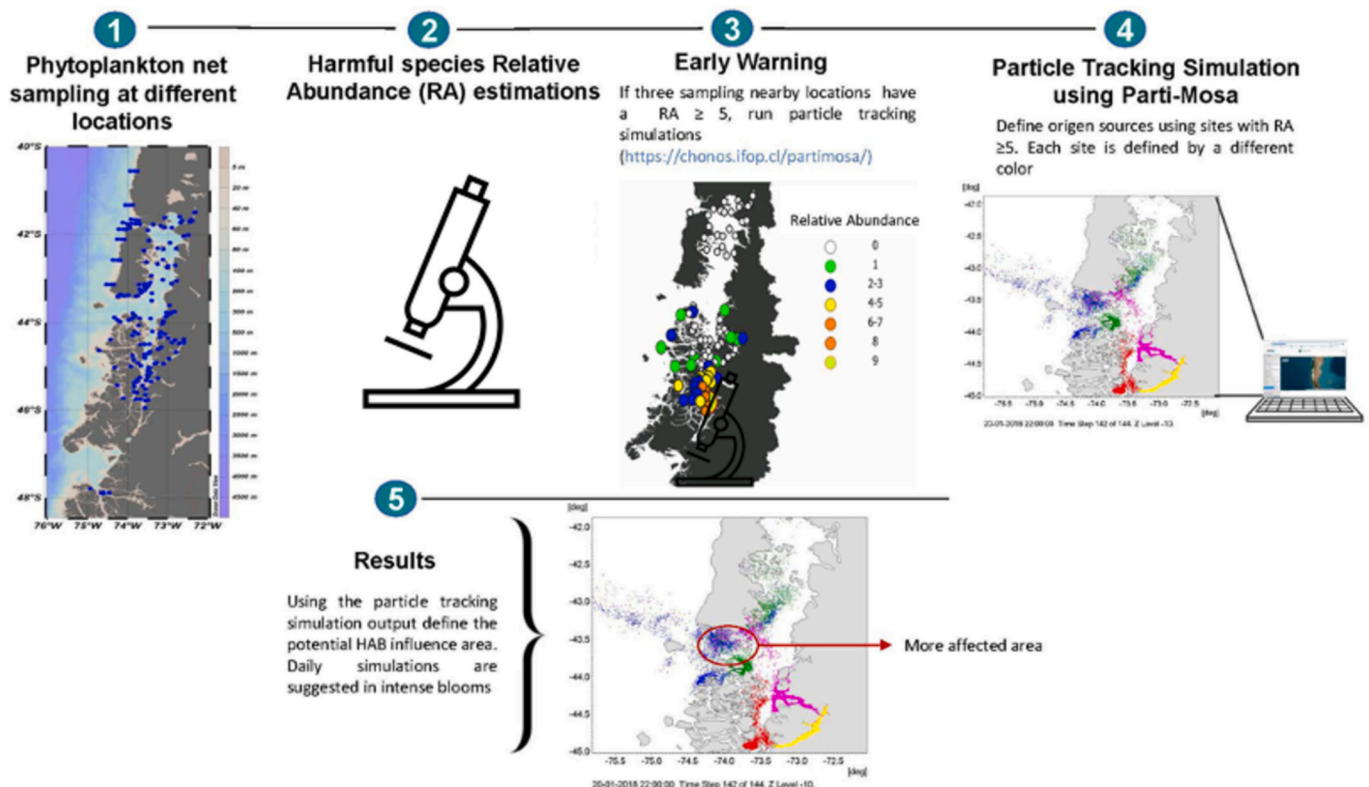


Fig. 1. Schematic explaining the basic steps of the Parti-MOSA model.

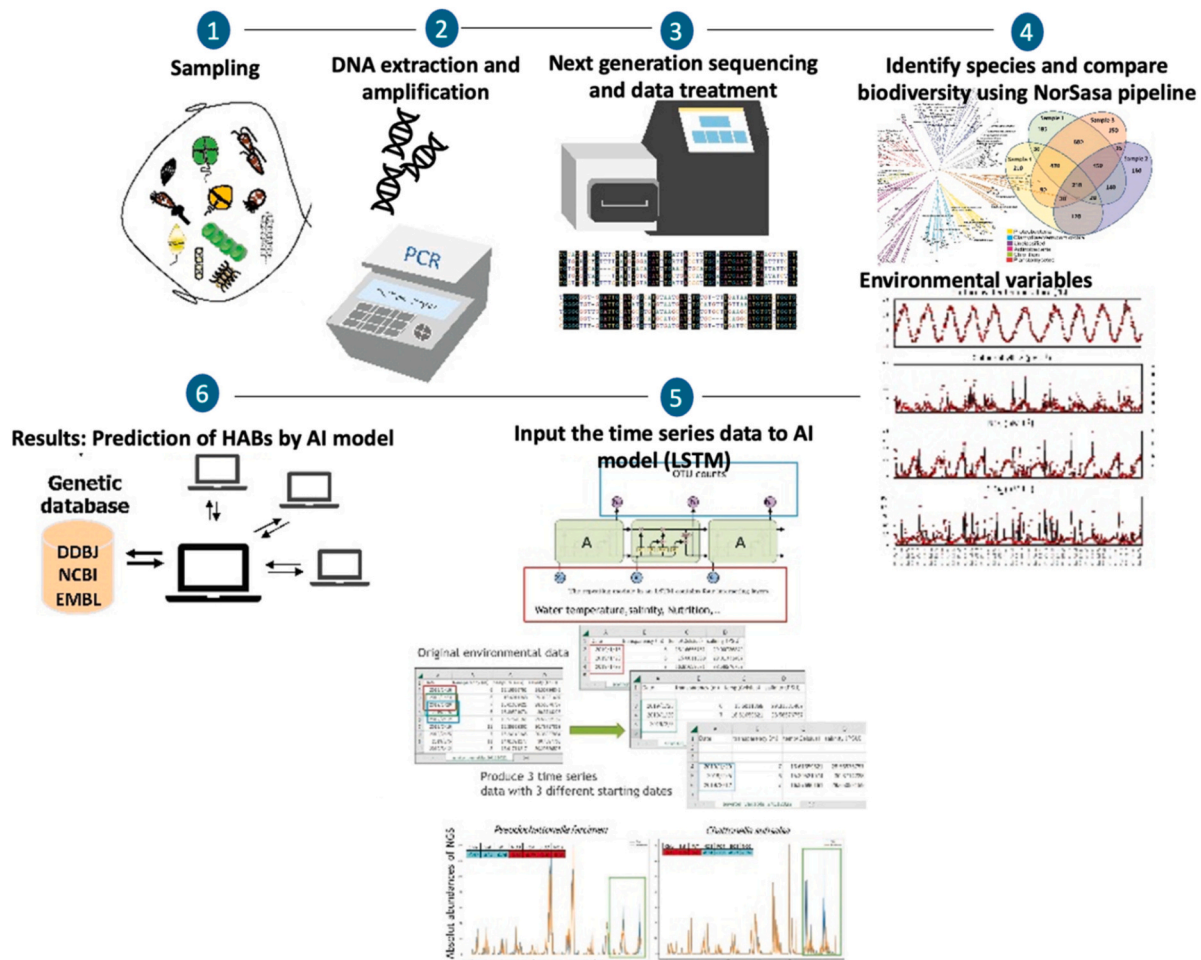


Fig. 2. Schematic explaining the basic steps of the LSTM model.

As such, EDM offers a novel approach to identifying causal interactions and generating predictions within HAB systems. The data and methods used in the case study are described below, and the results of applying the EDM model to HAB prediction in Chilean waters are evaluated (Fig. 3).

4. Methods

The EDM framework was applied to long-term phytoplankton monitoring data collected from three sites in Chilean Patagonia. Data processing and filtering were conducted to define target HAB species and potential predictor species (Section 4.1). Subsequently, convergent cross mapping (CCM) analyses were performed to identify causal relationships, which were then used to construct predictive models for HAB occurrences (Section 4.2).

4.1. Sampling sites and target HAB species selection

A 30-year phytoplankton dataset was obtained from INTESAL, which has operated the Phytoplankton Monitoring Program (PROMOFI) since the *Heterosigma akashiwo* bloom event in 1988. PROMOFI monitors phytoplankton abundance and composition in the inland sea of Los Lagos and Magallanes regions (Chile) to assist the salmon industry in assessing the risk of fish mortality during HAB events. Sampling is conducted weekly during spring, summer, and fall at salmon farms, from the surface to a depth of 15 m. Algal cell counts are conducted using Sedgwick Rafter chambers and standard inverted microscopy.

From the various PROMOFI monitoring sites, three were selected for

this study: Metri (41°35'S; 72°42'W), Quellón (43°07'S; 73°37'W), and Melinka (43°53'S; 73°44'W), all located in Northern Chilean Patagonia. These sites were chosen because of their distinct geophysical characteristics, including fjord systems with varying freshwater influence and circulation patterns. Metri is a site influenced by freshwater input (close to the discharge of the Puelo river, the most important river in that area), in contrast to Quellón and Melinka, which have a more oceanic influence and limited interaction with freshwater. Such differences are expected to result in site-specific bloom dynamics, suggesting that the effects of phytoplankton assemblages and physicochemical parameters on HAB species may vary by location.

During the selection of target species for EDM analysis, sporadic species were excluded because the algorithm in the current EDM framework could not capture variations in rare species. This screening process identified two common HAB taxonomic groups, *Pseudo-nitzschia seriata* and *Pseudo-nitzschia delicatissima*, which were selected as the primary targets for predictive modeling. More detailed information about the sampling sites, environmental characteristics, and the rationale for selecting these specific *Pseudo-nitzschia* species groups is provided in Perera et al. (2025). For the purpose of forecasting HABs using EDM, the use of *Pseudo-nitzschia* species groups remains valid, as the majority of the species comprising the two groups are known domoic acid producers based on their distribution and abundance.

4.2. Statistical analysis

The time-series datasets from the three sampling areas were pre-processed to ensure consistent biweekly intervals. For weekly samples,

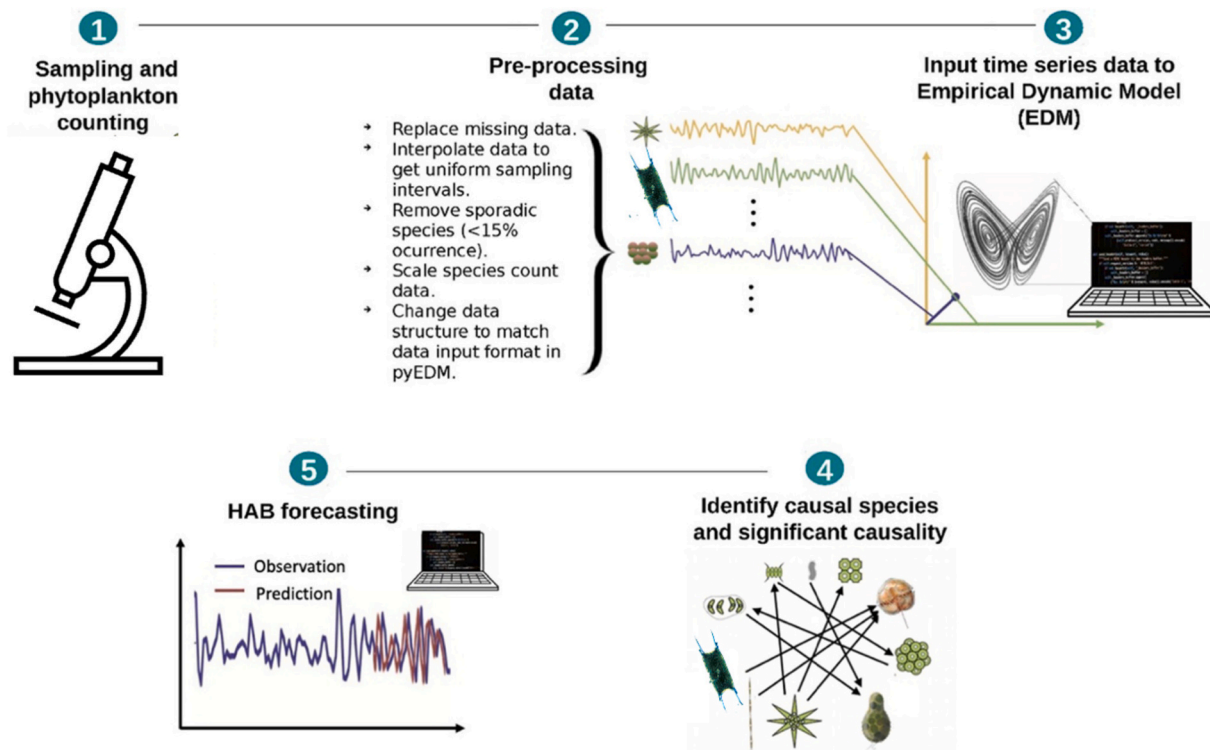


Fig. 3. Schematic explaining the basic steps of the EDM model.

data from the first and third weeks of each month were selected. The data period used in the EDM analysis was 2004–2020 for Melinka and 1992–2020 for Quellón and Metri. Although the PROMOFI program began collecting phytoplankton data at Melinka in 1999, the first five years showed inconsistent sampling and were excluded from the analysis. Phytoplankton abundance was calculated as total cell count per day. Only frequently occurring phytoplankton species, those present in more than 15% of the sampling dates, were included in the analysis.

convergent cross mapping was employed to estimate causal relationships between the two *Pseudo-nitzschia* species groups and other phytoplankton species. Prediction accuracy was evaluated using the Pearson correlation coefficient (ρ) between observed and predicted time series. This process, known as cross-mapping, assesses whether increasing the library size (i.e., the number of time points used to reconstruct the attractor; 200 points in this study) leads to convergence in prediction accuracy, a key indicator of causality.

The phase-locked twin surrogate method was used to generate surrogate time series that preserve seasonal patterns but lack causal relationships. Cross-mapping coefficients from these surrogate series provided a confidence interval to test for significant deterministic causality in the original time series (Ushio et al., 2018).

The CCM analysis followed standard procedures. Time series were normalized to a mean of zero and a standard deviation of one. The embedding dimension (E) and time delay were determined using the Simplex Projection method (Sugihara and May, 1990). Because CCM results are sensitive to the choice of E—the number of time-lag coordinates used to reconstruct system dynamics—E was selected based on the value that maximized predictability while avoiding overfitting (Ushio et al., 2018). Predictability was measured using correlation (ρ) between observed and predicted values.

Cross-mappings were used to estimate causality between variables, with a significance set at $p < 0.05$ compared to 100 phase-lock surrogate data. To examine the variation in causality strength along the temporal gradient, a multivariate S-map (sequential locally weighted global linear map; Deyle et al., 2016) was employed. The multivariate S-map quantifies the temporal variation in causality strength by conducting locally

weighted cross-mapping using all points within the multivariate dynamical system, rather than just neighboring points.

Causality strength was examined only for phytoplankton species that showed clear causality in the CCM graphs. The S-map was classified as positive ($\geq 90\%$ of values positive), negative ($\geq 90\%$ of values negative), or mixed (both positive and negative values). A species was considered to have a causal effect on the HAB if (1) CCM ρ values increased and plateaued with library size (convergence), and (2) the final ρ exceeded the 95% confidence interval of the surrogate data.

From the significant causal species identified by CCM, those with consistently positive causal influence ($\geq 90\%$ positive S-map coefficients over time) were selected as predictor variables for the HAB forecast model. Using these causal phytoplankton species, the two HAB species were predicted via the multivariate S-map method. Terms and methods described here are detailed in Munch et al. (2020) and Chang et al. (2017) (Supp. Fig. 2). Note that the predictions were made for the existing time series rather than a separate test set, to assess how well the causal variables could reconstruct the target series. Prediction performance was evaluated using the Pearson's correlation coefficient and associated p -values.

All analyses were carried out using Python version 3.9.7 and the pyEDM package version 1.13.1. The Python code used for analysis could be provided on request.

5. Results

5.1. Causality between the two *Pseudo-nitzschia* species and other phytoplankton species

CCM graphs combined with phase-lock surrogate analysis were used to identify phytoplankton species that potentially exhibit causal interactions with the two *Pseudo-nitzschia* groups (Fig. 4). In the graph, the red line (with a pink-shaded 95% confidence interval based on surrogate data) represents the ability of a phytoplankton species to predict the *Pseudo-nitzschia* time series, while the blue line (with a turquoise band) indicates the predictive skill of *Pseudo-nitzschia* for the other species. The

analysis primarily focused on the direction from other phytoplankton → *Pseudo-nitzschia*. A causal effect was considered significant if the red line exceeded the pink-shaded region, indicating that cross-mapping skill surpassed the 95th percentile of surrogate-based predictions.

Among the species showing significant causal influence, those with consistently positive interaction strengths across the study period were

identified using the S-map method and included in subsequent predictive modeling. The genus *Ceratium* and *Leptocylindrus* were common causal genera for both *P. seriata* and *P. delicatissima*. Other significant causal species with positive interactions varied between the two *Pseudo-nitzschia* species groups and across the three sampling sites and were predominantly diatoms.

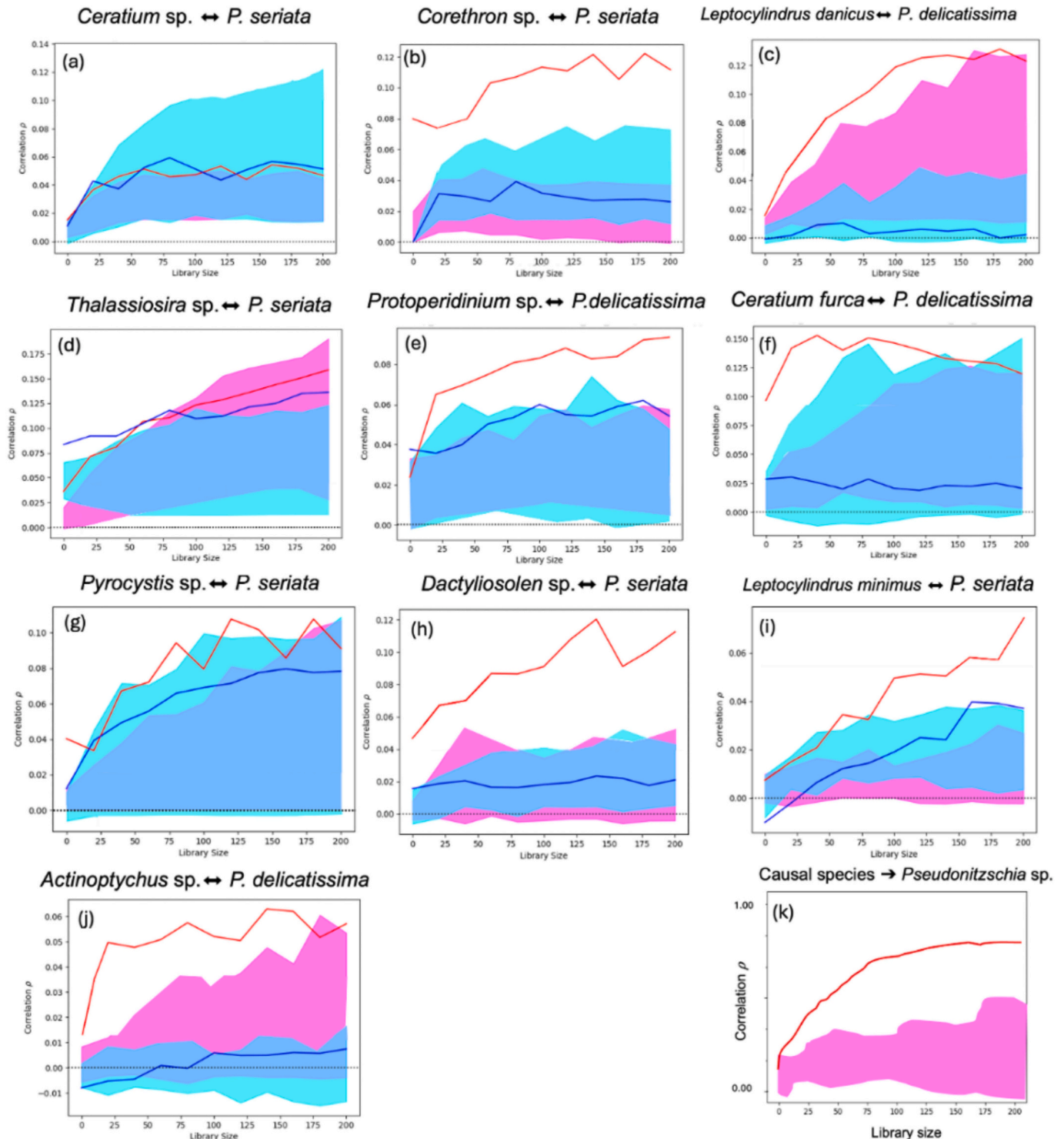


Fig. 4. CCM graphs with surrogate analysis depicting the causality of other phytoplankton species on *Pseudo-nitzschia* sp. (Red line and pink shaded area) and vice-versa (Blue line and turquoise shaded area). In this study we only focus on the redline (Actual correlation coefficient) and the pink shaded area (Surrogate data). Subplots (a-c) show results for Quellón, (d-f) for Melinka, and (g-j) for Metri. The plot at the lower right corner (k) is a depiction of a CCM graph with perfect causality from other phytoplankton to the target HAB species. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

5.2. *Pseudo-nitzschia* species prediction using causal species

The EDM showed variable predictive skill for the two *Pseudo-nitzschia* species groups across sites. The embedding dimension (E) ranged from 2 to 6, reflecting differences in system complexity between species and locations. The highest predictive accuracy was observed for *P. seriata* in Metri ($r = 0.733$, $P < 0.0001$), with the lowest MAE (0.137) and RMSE (0.994). In contrast, forecasts for both *P. seriata* and *P. delicatissima* in Quellón and Melinka exhibited weaker correlations ($r = 0.055$ – 0.190) and higher error values, indicating limited predictive performance in these areas.

The models were generally effective in predicting bloom occurrence (presence/absence) but showed lower skill in forecasting quantitative abundance, as reflected by modest correlation coefficients and relatively high error metrics. In other words, the models could indicate whether a bloom was likely to happen but not reliably predict exact cell concentrations (Fig. 5; Supp. Tables 2–4).

6. Discussion

6.1. Ecological interpretation of causal species relationships

The majority of identified causal species were diatoms; a noteworthy finding given their tendency to flourish in high-nutrient bloom conditions and often co-occur during HAB events. Diatoms such as *Leptocylindrus* and *Thalassiosira* can rapidly proliferate during blooms due to their high growth rate and chain-forming behavior. Species such as *Leptocylindrus danicus*, *Leptocylindrus minimus*, and *Thalassiosira* sp. have previously been associated with bloom events in Chile, some of which have been harmful to farmed fish and marine life (Barría et al., 2022).

Ceratium spp., a genus of dinoflagellates, is known to thrive in turbulent, nutrient-rich waters, conditions that also favor many diatoms.

These species possess distinctive traits such as high swimming capabilities, the ability to form cysts, and potential mixotrophic behavior (Rojas-Castillo et al., 2023). Such characteristics suggest a broad ecological niche that may temporally overlap with *Pseudo-nitzschia* occurrences, indicating possible interactions that could influence bloom development.

These co-occurring blooms may affect each other's dynamics, for instance, through competitive or facilitative interactions, particularly during bloom decline (Smetacek, 1985). This raises the possibility of using one HAB species as an ecological sentinel for others; for example, detecting a *Pseudonitzschia* bloom could serve as an early warning for an impending bloom of another harmful species. Such insights could enhance practical monitoring and forecasting strategies by integrating species interactions alongside environmental drivers.

Despite ongoing efforts to understand ecological interactions among phytoplankton communities during HABs, species-species interactions remain poorly characterized, largely due to limited long-term datasets (García-Girón et al., 2020; Padisák et al., 2016; Perera et al., 2025). While this study used an empirical model to identify species with potential causal relationships, the underlying ecological mechanisms remain unclear. Each interaction is likely context-dependent and may vary by species and location. It remains to be determined whether these relationships are direct (e.g., allelopathy or competition) or indirect (e.g., responses to shared environmental drivers). Future research could integrate EDM with environmental variables to better resolve these mechanisms, providing a more predictive framework for HAB management and ecological risk assessment.

6.2. Causal inference methods for HAB prediction

Identifying causal variables and their linkages to a specific ecological phenomenon is a central goal in ecology. The wide range of methods

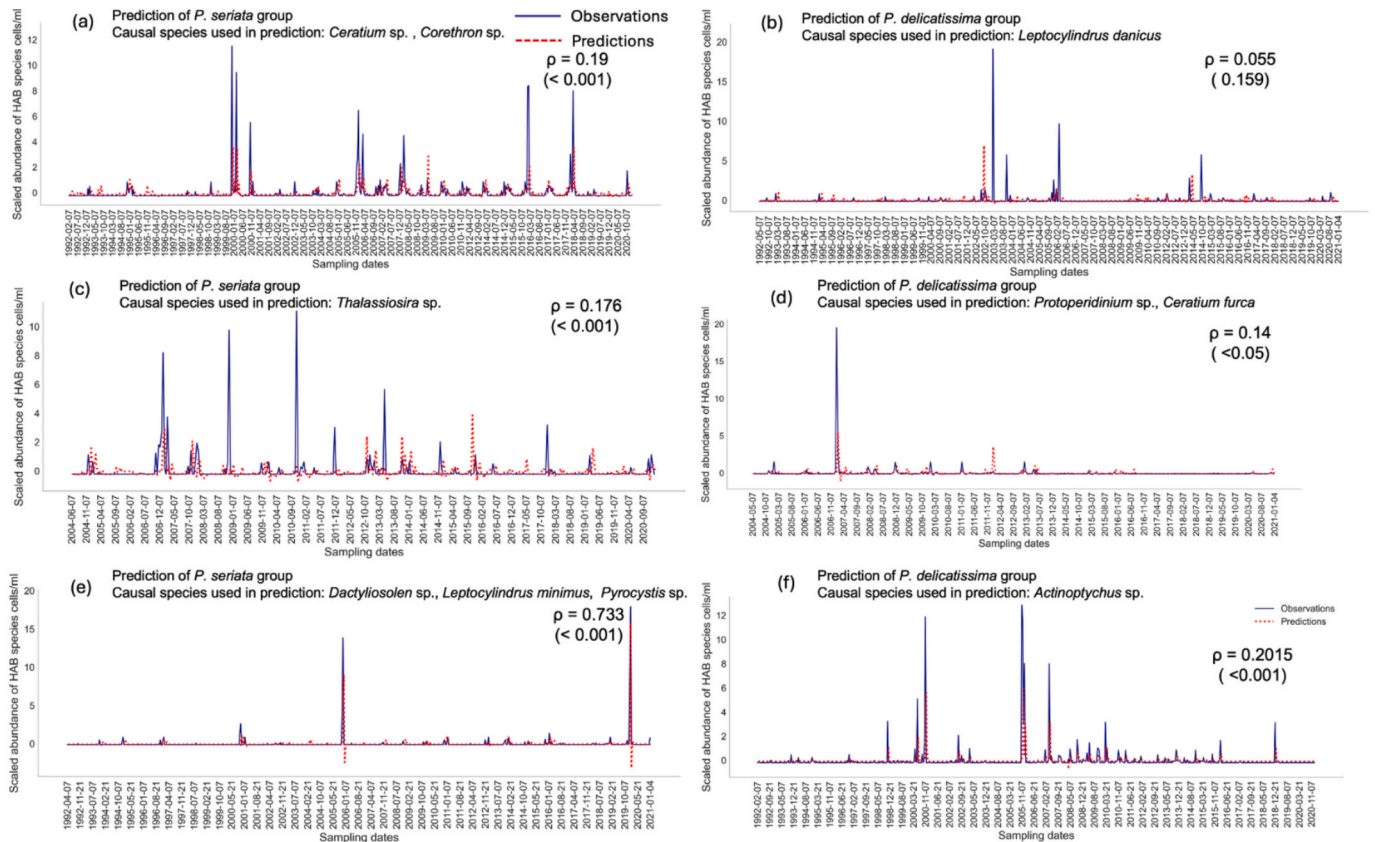


Fig. 5. The *Pseudo-nitzschia* species groups predicted using causal species by the multivariate S-map method. The accuracy of the prediction was evaluated by the Pearson correlation analysis and p values. Subplots (a, b) show results for Quellón, (c, d) for Melinka, and (e, f) for Metri.

proposed to infer causality from observational data reflects differing assumptions about system behavior, whether linear and stochastic, nonlinear and deterministic, or a combination of both. Each approach varies in suitability depending on the system and research domain. Methods based solely on correlation often struggle with nonlinear dynamic systems. As shown by Sugihara et al. (2012) and Munch et al. (2023), causally coupled variables may exhibit no consistent long-term correlation or may even show correlations that change sign over time, a phenomenon referred to as “mirage correlation.” These challenges make causal inference in ecological systems particularly difficult, especially when interactions are context-dependent or mediated by unmeasured environmental drivers.

Constructing detailed causal networks from time-series data, such as those shown in Fig. 4, remains complex. In this context, model-free approaches for inferring causality from observational data, such as EDM, offer promising opportunities. Unlike prior HAB forecasting studies that rely primarily on environmental variables or univariate species abundances, this study emphasizes species-species interactions as a core component of prediction. This approach represents a preliminary step toward evaluating whether causal species relationships among phytoplankton species can improve forecasts of HAB dynamics.

6.3. Performance of EDM-based predictions

The results demonstrate that the EDM can capture *Pseudo-nitzschia* bloom dynamics effectively under certain ecological conditions, particularly for the *P. seriata* group in Metri. Strong correlation and low prediction errors at this site suggest that local environmental and biotic factors are relatively deterministic, allowing EDM to reconstruct the underlying attractor accurately.

In contrast, lower predictive skill in Quellón and Melinka indicates either higher stochasticity or weaker coupling between environmental drivers and *Pseudo-nitzschia* population dynamics. Variation in embedding dimensions further suggests that bloom complexity differs among species and locations, likely due to distinct hydrodynamic conditions, nutrient regimes, or co-occurring taxa such as *Ceratium*.

Overall, these findings highlight spatial heterogeneity in HAB predictability and underscore the value of incorporating species

interactions and site-specific data into dynamic modeling frameworks. This suggests that the EDM-based species interaction models may be well-suited for early bloom detection but integrating them with environmental variables may improve abundance forecasting (Fig. 6).

Similar approaches have been applied elsewhere; for example, Tian et al. (2024) detected causality between algal density and environmental variables such as water temperature, river flow, water level, and nutrient levels in the Han River in China. These causal variables were used to construct multivariate prediction models including EDM, Random Forest, multiple linear regressions, and Gradient Boosting Machine models (Tian et al., 2024). While such studies demonstrate the value of combining EDM with ML for HAB prediction, this study differs in emphasizing species interactions as a core predictive component, highlighting its novelty.

These results suggest that integrating EDM-based species interaction models into operational monitoring programs could enhance early detection of HAB events and support targeted mitigation strategies. Future research could further couple species interactions with environmental drivers to refine predictive performance and generalize forecasting across sites and bloom types.

6.4. Limitations of the current EDM approach

This study has several important limitations that should be acknowledged. First, the performance of the EDM-based predictions was evaluated using in-sample reconstruction correlations, without out-of-sample testing. This approach may invite overfitting and optimistic bias, potentially overestimate the model's true predictive capability when applied to new, unseen data. Future work should incorporate cross-validation or independent test datasets to more rigorously assess model generalizability and forecasting performance.

Second, while the models demonstrated reasonable skill in predicting bloom occurrence (presence/absence), they showed limited accuracy in predicting bloom abundance or intensity. This limitation is critical for operational forecasting, as aquaculture managers and public health officials require not only early warning of potential blooms but also quantitative estimates of their magnitude to assess risk levels and implement mitigation measures. The moderate RMSE and MAE values

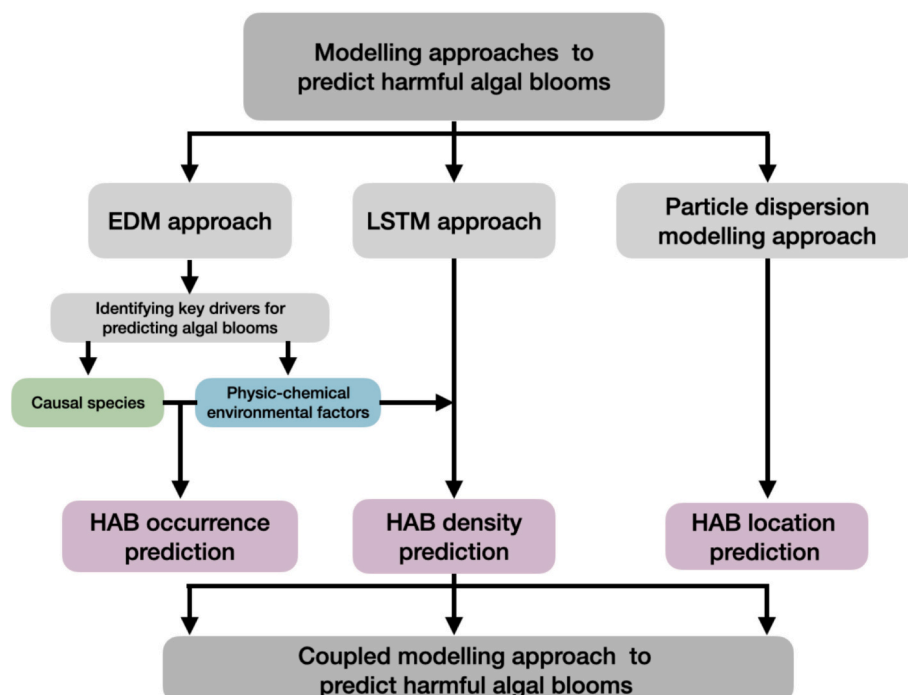


Fig. 6. Prototype approach for coupling the three SATREPS models.

observed across sites (Supp. Table 2) reflect this challenge.

Third, the current EDM approach focuses solely on species interactions and does not incorporate environmental variables, which are likely essential for improving predictions of bloom intensity. Prior studies have shown that factors such as salinity, upwelling, and temperature strongly influence *Pseudo-nitzschia* dynamics. (Anderson et al., 2009; Guallar et al., 2016; Lane et al., 2009; Palma et al., 2010; Perera et al., 2025; Vilas et al., 2014). Finally, site-specific variation in prediction accuracy suggests that model parameters and causal relationships may not transfer easily across regions, requiring location-specific calibration. This limits the immediate applicability of the approach to new monitoring sites without sufficient historical data.

6.5. Complementary approaches and future directions

Despite these limitations, the EDM-based species interaction approach provides valuable insights for HAB early warning systems, particularly for bloom detection. With advancements in real-time species identification and quantification—such as imaging devices integrated with AI algorithms—and integration with environmental and hydrodynamic models, EDM-based HAB prediction models could become increasingly accurate, timely, and actionable for operational forecasting. These findings also support a complementary approach to HAB forecasting, combining species interaction models with environmental variables to enhance predictive performance and generalize predictions across sites and bloom types.

7. Conclusion and recommendations

Models have become indispensable tools in HAB research, supporting activities from knowledge synthesis to predicting bloom events and informing management decisions. This study provided an overview of different HAB modeling approaches and presented a case study from Chile comparing three distinct methods: Parti-MOSA (physical particle-dispersion), LSTM (machine learning), and EDM (causal species interactions). The differences among these models highlight the potential value of coupled modeling approaches (Fig. 6). Such systems could combine the rapid response capability of Parti-MOSA, the pattern-recognition strengths of LSTM, and the ecological insight provided by EDM to deliver more robust and reliable HAB forecasts. The SATREPS-MACH project provides a foundation for implementing adaptive management strategies in Chile aimed at minimizing HAB impacts on public health, aquaculture, and fisheries. These advances would support Chile's ongoing efforts to develop data-driven, adaptive HAB management frameworks, offering a transferable model for other coastal regions facing similar challenges.

For operational implementation in Chile and elsewhere, coupled modeling systems should be designed with redundancy: if one model fails to detect a bloom (false negative), another might capture it; when multiple models agree, confidence in the alert increases, helping reduce both missed events and false positives. For instance, EDM could serve as a biological trigger for bloom occurrence, feeding into physical models like Parti-MOSA to map spatial dispersion. Our findings suggest that EDM performs particularly well in detecting bloom onset, even if less suited for quantifying bloom intensity. Combining it with environmental or oceanographic models could improve both timing and magnitude forecasts.

Future advances in HAB forecasting will depend on several technological and monitoring enhancements:

- 1) Real-time species identification and quantification: Integration of imaging devices with AI algorithms could enable rapid, automated phytoplankton identification, reducing the lag between sampling and prediction while improving data quality for EDM and LSTM models.

- 2) Enhanced spatiotemporal coverage: Expanding monitoring networks and increasing sampling frequency would improve model calibration and enable better detection of species interactions, particularly for less-studied HAB species.
- 3) Out-of-sample validation frameworks: Rigorous testing protocols using independent datasets or cross-validation approaches are essential to assess model generalizability and reduce overfitting bias identified in this study.
- 4) Integration of environmental sensors: Incorporating real-time oceanographic data (temperature, salinity, nutrients, upwelling indices) into coupled frameworks would address current limitations in abundance prediction.

This work contributes to the broader field of ecological informatics by enhancing HAB prediction through combined modeling approaches, grounded in both biological interactions and physical oceanography, that can lead to more effective, proactive management strategies. This integration has the potential to significantly reduce economic losses in aquaculture and better protect public health. This study takes a step in that direction by illustrating the value of coupled modeling for future HAB forecasting systems, contributing methodological advances that can be applied to diverse coastal ecosystems globally.

CRedit authorship contribution statement

Ishara Uhanie Perera: Writing – original draft, Formal analysis, Data curation. **So Fujiyoshi:** Methodology, Conceptualization. **Daiki Kumakura:** Methodology, Formal analysis. **Carolina Medel:** Writing – review & editing, Formal analysis. **Kyoko Yarimizu:** Writing – review & editing. **Osvaldo Artal:** Methodology. **Pablo Reche:** Methodology. **Oscar Espinoza-González:** Writing – review & editing, Methodology, Conceptualization. **Leonardo Guzman:** Writing – review & editing, Methodology, Conceptualization. **Felipe Tucua:** Methodology, Data curation. **Alexander Jaramillo-Torres:** Methodology, Data curation. **Jacqueline J. Acuña:** Writing – review & editing. **Milko A. Jorquera:** Writing – review & editing, Project administration, Methodology. **Shinji Nakaoka:** Supervision, Methodology. **Satoshi Nagai:** Writing – review & editing, Methodology, Conceptualization. **Fumito Maruyama:** Writing – review & editing, Project administration, Methodology, Conceptualization.

Funding sources

This work was supported by the Japanese Society for the Promotion of Science under Grants-in-Aid for Scientific Research (grant numbers 18K19674, 18KK0436, 20H00562, 20K18903), Science and Technology Research Partnership for Sustainable Development (grant number JPMJSA1705) and Osvaldo Artal was partially funded by COPAS ANID FB210021.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Fumito Maruyama reports financial support was provided by Japanese Society for the Promotion of Science. So Fujiyoshi reports financial support was provided by Japanese Society for the Promotion of Science. Fumito Maruyama reports financial support was provided by Science and Technology Research Partnership for Sustainable Development. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We thank Dr. Mikihiro Kawai of Kyoto University for providing

valuable technical suggestions during our monthly meetings with IFOP. We also thank Edanz (<https://www.edanz.com/ac>) for editing the English text of a draft of this manuscript. We are thankful to Héctor H. Sepúlveda for his valuable help in the development, implementation, and support of Parti-MOSA. We thank the Instituto Tecnológico del Salmón (INTESAL) for providing the long-term time series data from the Phytoplankton Monitoring Programme (PROMOFI), and acknowledge the database management team and the technical staff involved in the maintenance of this resource.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ecoinf.2026.103615>.

Data availability

The code and data supporting this study are publicly available at https://github.com/UhP88/HAB_EDM

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